

Qatar-2b

R-0977

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_260619 · created 2026-07-29T19:21:07+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461210.7365 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.223 +/- 0.064
Transit depth (Rp/Rs)^2	4.98 +/- 2.86 [%]
Orbital Inclination (inc)	89.53 +/- 1.74
Airmass coefficient 1 (a1)	0.71 +/- 0.44
Airmass coefficient 2 (a2)	0.18 +/- 0.26
Scatter in the residuals of the lightcurve fit is	61.6 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	3.0
Transit Duration (day)	0.0801 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.223 ± 0.064 vs 0.16327 (deviation 0.85σ)
Duration fit vs published	0.0801 d vs 0.0767 d (deviation 0.66σ)
Mid-time O–C	-5.0 min
Depth SNR	3.2
Residual scatter	61.6 %

Light curve

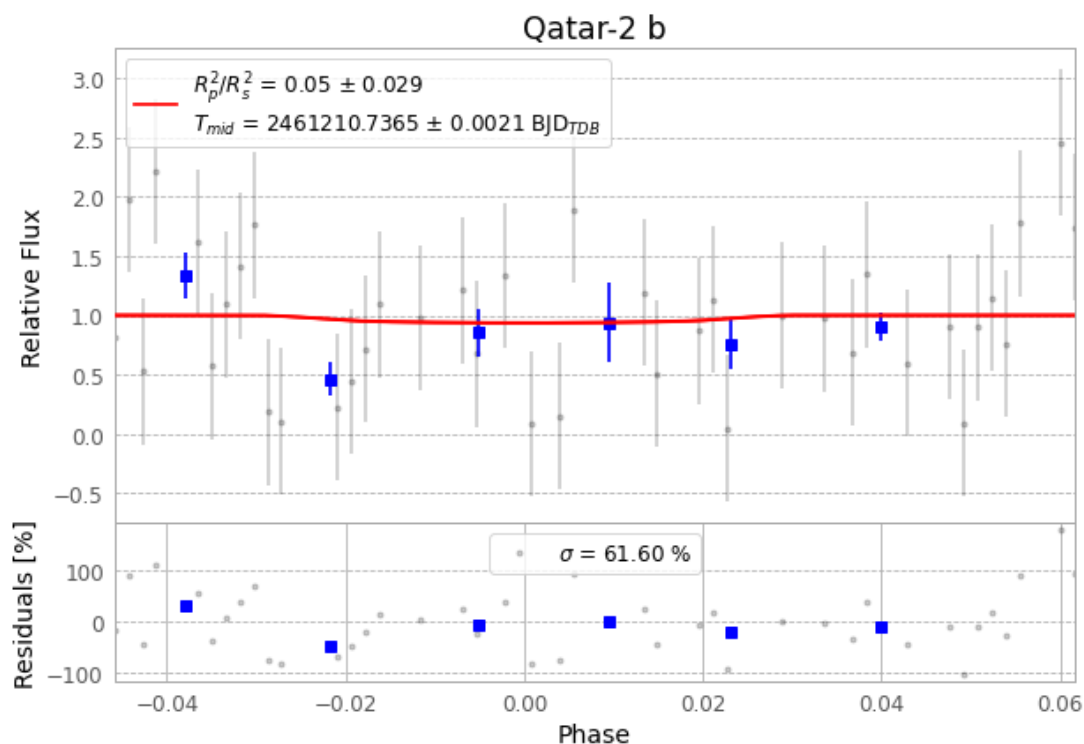


Figure 1 — FinalLightCurve_Qatar-2 b_2026-06-18.png (SHA-256 5a544b3d815ff6fe...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [614, 476], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 12082c0340af0ce5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

71 FITS frames (47 MB), Qatar-2260619040616.FITS → Qatar-2260619073616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-260619.log (5d7e8c088808...), FinalLightCurve_Qatar-2 b_2026-06-18.png (5a544b3d815f...), FinalLightCurve_Qatar-2 b_2026-06-18.csv (0646231a157f...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 19:21Z.